

Experiment 7: Breadth First Search (BFS)

Aim

To write a Python program to implement Breadth First Search (BFS) traversal on a graph.

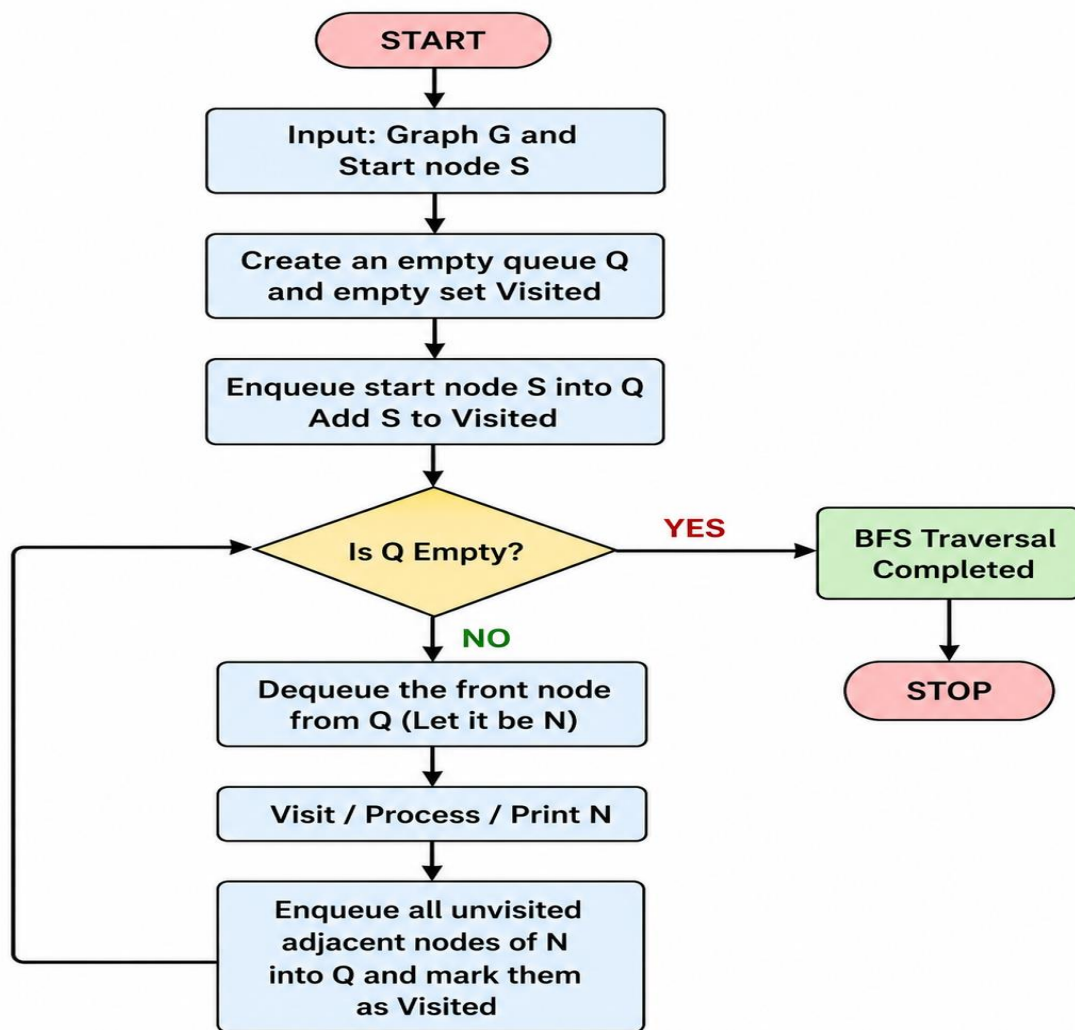
Objective

To traverse all the nodes of a graph level by level using the BFS algorithm and understand the working of queue-based graph traversal.

Algorithm

- Create a graph using an adjacency list.
- Initialize a queue with the starting node.
- Mark the starting node as visited.
- Remove a node from the queue.
- Print the node.
- Add all unvisited adjacent nodes to the queue.
- Repeat steps 4–6 until the queue becomes empty.

FLOWCHART:



Code

Python

Run

```
from collections import deque
```

```
g = {'A':['B','C'], 'B':['D'], 'C':['E'], 'D':[], 'E':[]}
```

```
q = deque(['A'])
```

```
v = set(['A'])
```

```
print("BFS Traversal:")
```

```

while q:
    n = q.popleft()
    print(n, end=' ')
    for i in g[n]:
        if i not in v:
            v.add(i)
            q.append(i)

```

Output

```

IDLE Shell 3.12.4
File Edit Shell Debug Options Window Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:30:16) [MSC v.1940 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/sandh/OneDrive/Desktop/EXPERIMENT 7.py
BFS Traversal:
A B C D E F
>>>
Ln: 7 Col: 0

```

Result

The BFS traversal of the given graph was successfully performed using Python, and the nodes were visited in the order: **A → B → C → D → E**.

Conclusion

The Breadth First Search algorithm was implemented successfully in Python. BFS visits graph nodes level by level using a queue and is useful for shortest-path and graph traversal applications.

ChatGPT can make mistakes. Check important info.